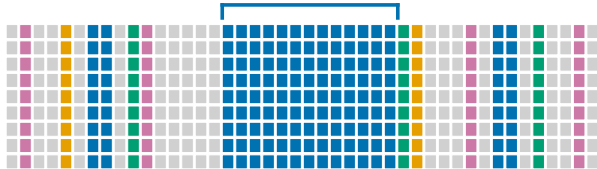


(a) Four-way acquired k-space partition

$$\Omega_v = \Theta_v \cup \Lambda_{\text{rec},v} \cup \Lambda_{\text{risk},v} \cup \Lambda_{\text{hold},v}$$

pairwise disjoint acquired subsets

ACS lines in Θ_v



- Θ_v — reconstruction input
- $\Lambda_{\text{rec},v}$ — SSDU reconstruction loss
- $\Lambda_{\text{risk},v}$ — residual-risk learning
- $\Lambda_{\text{hold},v}$ — independent verification
- unsampled k -space

Role assignment is fixed per acquired phase-encode line.
ACS lines are always assigned to Θ_v .

(b) Role-separated verification workflow

